NUCLEAR MAGNETISM

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A NEW field of research in physics has been opened within the last few years which is now commonly referred to as the study of "nuclear magnetism." This somewhat unusual combination of two apparently disconnected words deserves some explanation.

The word "nuclear" refers to the nucleus of the atom, the central part where most of the mass is concentrated. The second word, "magnetism," strikes a more familiar note in so far as magnetic properties of matter have been known for a long time and are now part of everyday experience. It may be well, nevertheless, to summarize briefly the basic facts and explanations of this normal and well-known type of magnetism in order to prepare the ground for the understanding of the new type with which this paper will be concerned.

As one of the most fundamental facts, it must be recalled that the properties of any magnetized substance, for example those of an iron magnet, are the result of a very large number of very small magnets in the interior of the substance. This fact rests upon the observation that each of the fragments obtained in breaking up the substance has in itself the properties of a magnet; the process of subdivision can, in principle, be carried on until one arrives at the "elementary magnets" which are evidently nothing else but the atoms or molecules which constitute all matter. Thus the existence of the elementary magnets remains to be explained.

The essential features of this explanation were given over a hundred years ago by Ampere who showed that a loop of wire, carrying an electric current, exerts at a distance the same forces of attraction as a magnet. By analogy he postulated that the behavior of atoms and molecules as elementary magnets would likewise have to be explained by the existence of minute electrical currents circulating in their interior. While this postulate has been widely accepted, it required the understanding of atomic structures, reached within the last forty years, in order to explain fully the nature of the "amperian currents." Figure 1 is a symbolical presentation of our present ideas about the atom. The central part represents the massive nucleus. The circular and elliptic curves around it symbolize the orbits of electrons with a mass which is several thousand times smaller than that of the nucleus. Each electron carries a negative electric charge, and the motion of these charges, like that in a wire loop, is equivalent to the existence of electrical currents. They represent, in fact, the amperian currents which give to the atom the

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property of an elementary magnet and, hence, must be recognized as the underlying cause of ordinary magnetism.

Basic Features of Nuclear Magnetism

Turning now to nuclear magnetism, we are dealing with elementary magnets of a different nature. In contrast to ordinary or "atomic" magnetism, nuclear magnetism is caused by amperian currents within the nucleus, and it is therefore necessary to outline briefly its origin. Unfortunately, there is at present not nearly as much known about the structure of nuclei as there is about that of atoms, and it is indeed one of the chief concerns of contemporary physics to gain more insight of this subject. While Figure 2, representing a nucleus, must therefore be considered as even more symbolic than Figure 1, it contains features which are

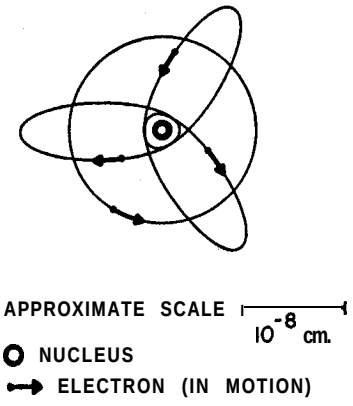


FIG. 1. Schematic presentation of an atom. The relative size of the nucleus is greatly exaggerated. In the same scale it would appeal about 10,000 times smaller than indicated here.

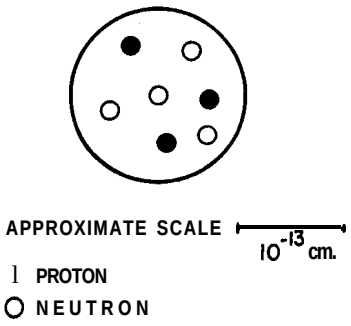


FIG. 2. Schematic presentation of a nucleus. Neutrons and protons must both be thought of in a state of rapid motion, confined within a region which is indicated by the outer circle.

based upon some well-established facts. In the first place, it has been understood that nuclei are by no means simple systems but that they themselves are composed of several particles termed the nucleons. Two kinds of nucleons are known: the proton, with a mass about two thousand times larger than that of the electron and carrying a positive electric charge; and the neutron with a mass almost, equal to that of the proton mass but electrically neutral, i.e., without a charge. These particles must be thought of as undergoing a rapid motion within a narrow region of spase, schematically indicated by the surrounding circle of Figure 2. The details of this motion remain to be investigated, but one of its features is well known and is of primary importance at this point: whatever the paths of the individual nucleons may be, it appears that, in about half of the known nuclei, there exists a rotation of the

nucleus as a whole about an axis, which passes through its center of gravity.

It must be noted that, as a rotating mass, the nucleus possesses the mechanical property of an angular momentum, directed along the axis of rotation. According to the laws of quantum mechanics, the magnitude of this angular momentum can only assume the values (0, ½, 1, 3/2, etc.) $h/2\pi$, where h is Planck's quantum of action and where the bracketed values indicate the "spin" of the nucleus. For all values of the spin, different from zero, i.e., in all cases where one deals with a finite rotation, there exists another property arising from the fact that not only the masses but also the electrical charges located in the nucleus participate in the rotation. This rotation of the charges, similar to the circulation of the electrons in the atom, is again equivalent to an amperian current so that nuclei can likewise be expected to have the properties of elementary magnets.

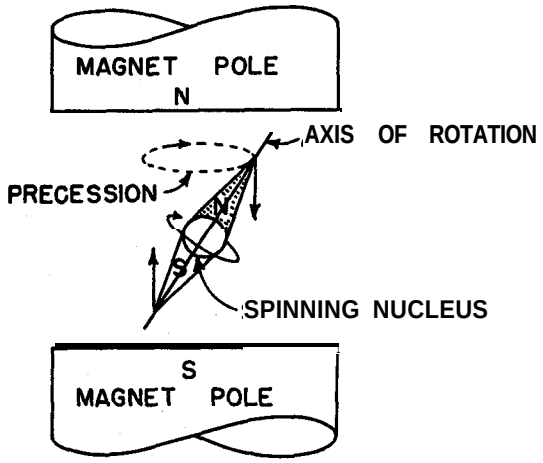


FIG. 3. Principal elements of nuclear precession. The drawing illustrates the precession of a nucleus. It is due to the torque arising from the action of the magnet upon the magnetic moment of the nucleus. The latter is symbolized in the figure by a compass needle, oriented along the axis of rotation.

While there is thus a qualitative similarity between atomic and nuclear magnetism, there exists at the same time a great quantitative difference in regard to the strength of the elementary magnets. It is customary to measure this strength by a quantity called the "magnetic moment" which, for a macroscopic bar magnet, is defined as the pole strength of the north and south pole, multiplied by their separation. Comparing an atom and a nucleus with the same angular momentum, it can be seen that their magnetic moments must be in the approximate inverse ratio of the masses of electrons and nucleons, respectively, and it is an observed fact that the magnetic moment of a spinning nucleus is about a thousand times smaller than that of an atom.

In describing the origin of nuclear magnetism, we have restricted ourselves not only to its purely qualitative aspects but we have also omitted another essential fact. It would seem, from this description, that the motion of the nucleons is a necessary requirement for the existence of a nuclear magnetic moment and that a proton at rest could not contribute to the magnetic moment of a nucleus. Actually this is not the case, since the proton itself is known to possess a spin ½ and a magnetic

moment. This "intrinsic" spin of an elementary particle may in some respects be thought of as a rotation around an axis which passes through the particle itself, somewhat analgy to the daily rotation of the earth. There is not much known, at present, about the corresponding amperian currents which evidently circulate within the particle and thus give rise to its intrinsic magnetic moment. Their somewhat peculiar origin is emphasized by the fact that the neutron, besides having a spin $\frac{1}{2}$, likewise exhibits a magnetic moment in spite of the fact that it carries no total electric charge. A quantitative understanding of nuclear magnetic moments requires that the intrinsic moments of the nucleons be taken into consideration. Furthermore, the simplest of all atoms, that of hydrogen, has a nucleus which consists of a single proton so that one deals in this case altogether with properties arising from the intrinsic spin and magnetic moment of a nucleon. The nucleus of the hydrogen atom is of particular interest, here since a great, deal of the study of nuclear magnetism has until now been carried out on matter containing hydrogen.

New Methods of Study

The fact that the elementary magnet's of nuclear magnetism are very much weaker than those of atomic magnetism necessitates new and different methods of study. In order to explain these methods, we shall begin by considering a single nucleus, located between the poles of a laboratory magnet. The essential features of the situation are illustrated in Figure 3. Because of its magnetic moment, the spinning nucleus behaves as if a compass needle were rigidly attached to it and oriented in the direction of the axis of rotation. The north pole (N) of this equivalent compass needle will be repelled from the north pole of the magnet by a downward force, and the south pole (S) will be repelled from the south pole of the magnet by an upward force. These two equal and opposite forces result, in a torque upon the nucleus. The effect of this torque is illustrated in Figure 4 by the analogous situation of a spinning top where the downward force is provided by the weight of the top and the upward force by the equal and opposite reaction acting at

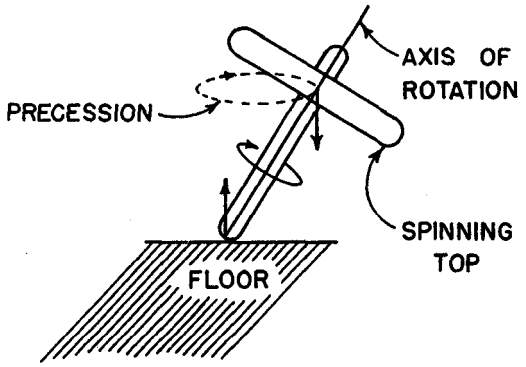


FIG. 4. Precession of a spinning top. The drawing illustrates the analogous situation of a spinning top. The torque qrises here from the gravitational force, acting upon thr center of mass of the top and from the equal and opposite force of reaction acting at the point of contact, with the supporting floor.

the point of contact with the supporting floor. Common experience shows that the axis of the top carries out a slow rotation around the vertical. This rotation, to be distinguished from the rotation around the axis, is called "precession," and by analogy we have to expect that the axis of the spinning nucleus, and hence its magnetic moment, performs a similar precession.

The rate at which this precession occurs is proportional to the strength of the laboratory magnet or, more accurately, to the magnitude of the magnetic field which exists at the location of the nucleus. The factor of proportionality is given by a quantity which is characteristic of the nucleus; it is called its "gyromagnetic ratio" and is defined as the ratio of the magnetic moment to the angular momentum of the nucleus. If the magnetic field has the easily attainable magnitude of, say, a thousand oersted and if we use as an exemplifying nucleus that of hydrogen, it is found that the precession occurs at the rate of about four million cycles per second.

This frequency, although much higher than that of a macroscopic top, is actually very low in comparison to the frequency of motion of the nucleons within the nucleus. It must be noted, however, that it lies in the range of radio frequencies-a fact which is of importance for the experimental techniques.

The experiments are, of course, not carried out on a single nucleus but on the very large number of nuclei contained in a macroscopic sample. The observation of protons, for example, requires the presence of an appreciable amount of hydrogen ; one cubic centimeter of water in a test tube may serve to illustrate the practice.

The method of observation is schematically indicated in Figure 5. The orientation of the magnet poles has been rotated with respect to Figure 3 so that the magnetic field in the gap between the poles has a horizontal instead of a vertical direction. The purpose of this field is twofold. In the first place, it has the effect of causing a slight nuclear magnetization of the sample. Indeed, in the absence of a magnetic field one would have a random orientation of the magnetic moments of the nuclei so that their effects would cancel each other. Due to the previously discussed torque on each moment caused by the magnetic field, there occurs, however, a preferential orientation of the moments with the effect that a resultant magnetic moment, parallel to the magnetic field, is established. This "nuclear polarization" of the sample is very slight; while the corresponding polarization in the case of ordinary magnetism manifests itself by measurable forces exerted upon other magnetized bodies, one deals here with such small forces that their direct measurement would be very difficult.

The actual detection of nuclear magnetism makes use also of the second effect of the magnetic field which was previously discussed-the fact that

the magnetic moment of each nucleus is caused to perform a precession. Visualizing the total nuclear polarization as a compass needle, one has to imagine that its direction likewise performs a precession with the same frequency as that of the nuclei of which it is the resultant. It is true that there is no detectable effect as long as the polarization is oriented in a direction parallel to the magnetic field, since the precession occurs in this case with "zero angle" and leaves the direction of the polarization unchanged. To produce a detectable effect it is necessary to provide a mechanism by means of which the polarization is tilted by a finite angle with respect to the magnetic field. This can be done most easily by superimposing over the magnetic field, originating from the magnet poles, an oscillating field at right angles to it. In the drawing of Figure 5 the direction of this oscillating field is assumed to be perpendicular to the paper. Provided that the frequency of this oscillating field is equal to that of the nuclear precession, there occurs "magnetic resonance." an effect which is similar to mechanical resonance and which can be shown to produce the desired tilt of the nuclear polarization.

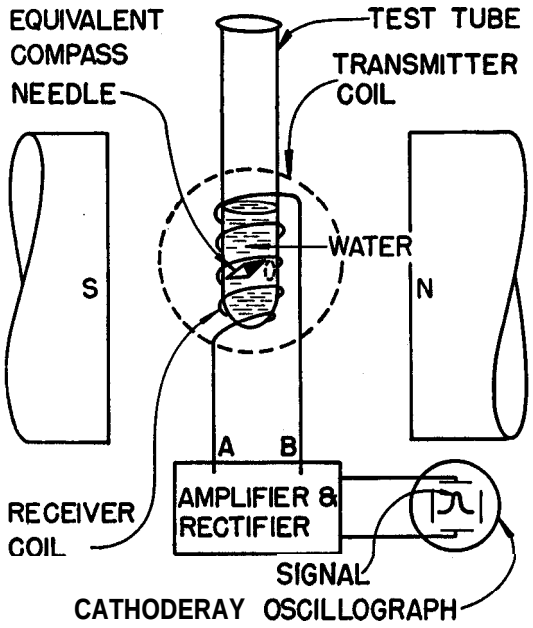


FIG. 5. Schematic scheme of observation. Under the influence of the field in the gap between the pole faces, the water in the test tube undergoes a slight nuclear magnetization and behaves in this respect like an equivalent compass needle, symbolically indicated in its interior. The adjoining small dotted curve represents the precession of this equivalent compass needle, occurring at the same rate as that of an individual nucleus, as shown in Figure 3. It causes an induced alternating voltage in the receiver coil, wound around the sample. After amplification and rectification, this voltage is displayed as a signal on the screen of the cathode-ray oscillograph.

The nuclear polarization, tilted, against the magnetic field and performing a precession around its direction, has the effect of an invisible compass needle rotating within the test substance--in this case a small amount of water. The phenomenon is indeed invisible in the sense that it is not related to visible light but this circumstance does not prevent its detection by other means. The detection is, in fact, based upon the well-known principle of electromagnetic induction, discovered long ago by Faraday who found that a change of magnetic: flux within a coil of wire causes a voltage difference between its terminals. To reproduce this situation it is merely necessary to wind a few turns of wire around

the sample as indicated in Figure 5 by the "receiver coil." The precession of the titled polarization induces an alternating voltage of the same frequency between the terminals *A, B* of this coil, and it is the observation of this voltage, i.e., the phenomenon of "nuclear induction," which forms the basis for the investigation of nuclear magnetism1

The experimental procedure may be compared to that followed in the reception of radio signals. It can be stated to consist of the reception of radio signals originating from the nuclei in the sample with the receiver coil serving the purpose of an antenna, a device to pick up the radio signals. The further steps are exactly those performed in an ordinary radio

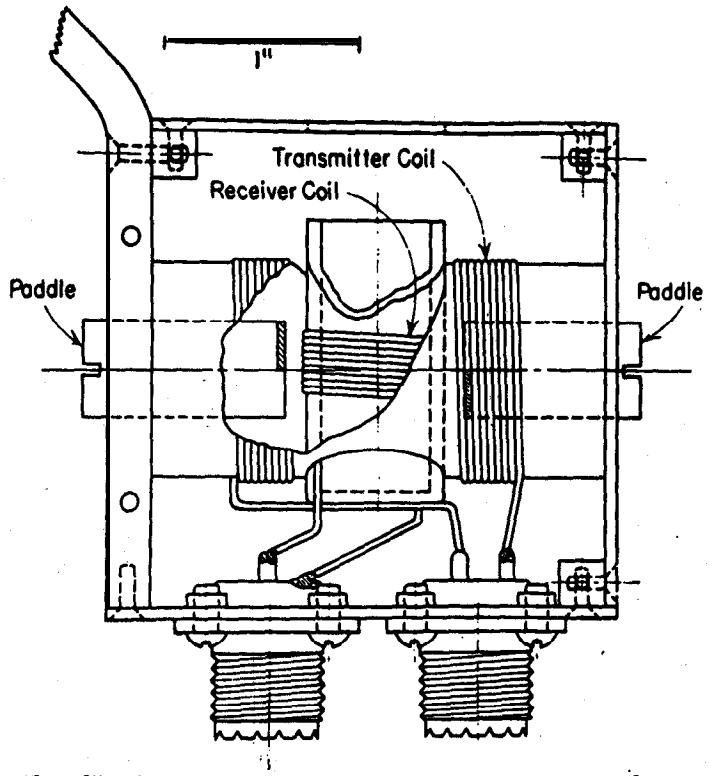


FIG. 6. The "head" of the assembly. Test tubes can be inserted into the cylinder, indicated at the center with the receiver coil wound around it. The "paddles" on the right and left can be rotated to achieve sufficient decoupling between the receiver and the transmitter coil.

receiver; they consist of the amplification and rectification of the nuclear induction signal, and in the ultimate operation of a loudspeaker. Instead of making the signals audible, however, it is more convenient to make them visible by displaying them on the screen of the cathode-ray oscillograph. This last stage of detection is schematically indicated in Figure 5 with a typical curve sketched on the screen, which represents a signal as it is traced out by the cathode ray under the conditions of magnetic resonance.

There is one more element which is essential for the observation of nuclear induction. It consists of another coil through which an alternating current is sent in order to produce the oscillating field, necessary for the

occurrence of magnetic resonance. This coil, to be distinguished from the receiver coil, is called the "transmitter coil"; its axis must be thought, in Figure 5, to be perpendicular to the paper in order to give the same direction to the oscillating field. The transmitter coil is usually made to consist of two halves in order to allow simple insertion of the test tube, containing the sample. In Figure 5 one has to think of one half being located above, the other half below, the paper and, viewed from above, they are both indicated by the dotted circle in the figure.

The following pictures illustrate the methods of studying nuclear magnetism which we have just discussed. Figure 6 is a drawing of the

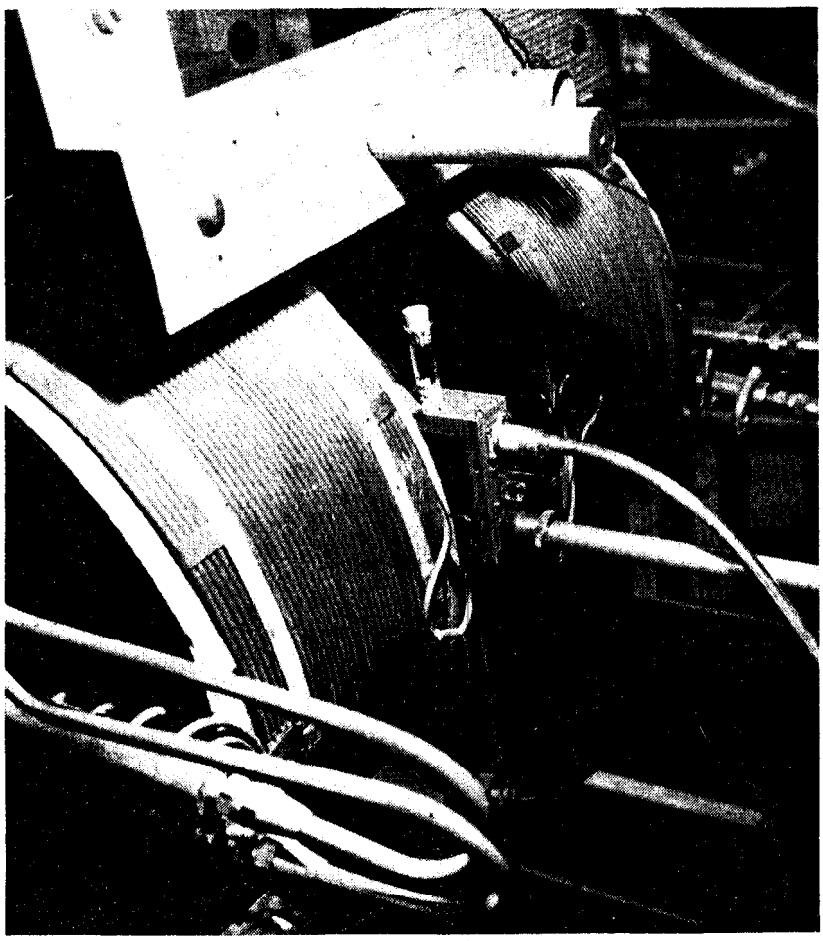


FIG. 7. The mounted head, carrying a test tube and ready to be inserted in the gap of the electromagnet.

"head," the central part of the assembly which is placed between the poles of the laboratory magnet so that the field here is perpendicular to the paper. The test tubes are inserted through a hole in the top of the head into an insulating cylinder surrounded by the receiver coil. The oscillating field, produced by an alternating current through both halves of the transmitter coil, points alternately from right to left and from left

to right and acts upon the nuclei of the test substance in the center. Care has to be taken in this arrangement that the change of magnetic flux, due to the oscillating field, does not directly induce a far greater voltage in the receiver coil than that due to nuclear induction, and thus overshadow the latter. This is partly achieved by mounting the receiver and the transmitter coils with their axes at right angles. A fine adjustment for the decoupling of the two coils is indicated by the two "paddles," drawn in the figure. They consist of insulating plugs which carry a semicircular copper disk at their inner ends. By a suitable rotation of the plugs one can steer the oscillating field into such a direction that the flux through the receiver coil is sufficiently reduced. In operation, there are cables attached to the two connectors at the bottom, one of which supplies the transmitter coil with an alternating current from a tank circuit while the other connects the receiver coil to the amplifier.

Figure 7 shows a head, carrying a test tube, as it is about to be introduced between the poles of an electromagnet, energized by the two large visible coils. The bottom rod which supports the head carries the cable to the transmitter coil. The bent cable, above the rod, leads from the receiver coil to the amplifier.

Figure 8 is a photograph of a nuclear induction signal from the protons in water in a field of about 2000 oersted, traced out on the screen of a cathode-ray tube. The vertical displacement measures the magnitude of the signal; the horizontal displacement is proportional to the strength of the magnetic field which is made to increase steadily with time. The value of the field, reached at the center and resulting in the maximum of the signal, is that at which the condition of resonance is fulfilled, that is, where the frequency of precession of the protons is equal to the frequency of the applied oscillating field. As the field deviates on both sides more and more from the resonance value, the signal becomes smaller and disappears when the two frequencies differ appreciably. The "half-width" of this resonance curve-the width measured between the two points where the signal has half its maximum value-corresponds to a field variation about one-half oersted.



FIG. 8. Photograph of an absorption signal from protons in water.

Figure 9 shows a signal obtained from the same sample as in Figure 8 with resonance occurring again at the center. The difference between the two presentations originates from a difference in the phase of the nuclear induction signal with respect to that of the oscillating field. In the terminology of optics, one deals in Figures 8 and 9 with two related phenomena, that of absorption and of dispersion, respectively. Either one or a combination of both can be obtained by the adjustment of the phase-sensitive detector.

We shall now turn to some of the results which have been obtained through the study of nuclear magnetism as described above.

Measurement of Nuclear Magnetic Moments

The most direct and, fundamentally, the most significant results are obtained in the measurement and comparison of nuclear magnetic moments. It was stated before that the magnetic moment of a nucleus is caused by the amperian current flowing in its interior. Conversely, it is possible from the knowledge of the magnetic moment to draw important, conclusions concerning the structure of a nucleus and the motion of its constituent particles. The evidence collected in this manner has greatly contributed towards revealing a "shell structure" of nuclei, somewhat analogous to the order of atomic structures which is manifested by the periodic system of the elements.

Once the condition of magnetic resonance has been established by the observation of a nuclear induction signal, one merely has to determine the magnetic field and the frequency of the oscillating field. These two data furnish directly the value of the gyromagnetic ratio and, if the spin is known, of the magnetic moment of the nucleus under investigation. Several independent, methods exist by which it is possible to determine the nuclear spin and, hence, the angular momentum so that a determination of the gyromagnetic ratio of a nucleus becomes equivalent to the measurement of its magnetic moment. But even in cases where such independent information is lacking, it can be derived by the observation of nuclear magnetism itself in so far as the magnitude of the observed signal furnishes an additional datum from which the spin can be computed.

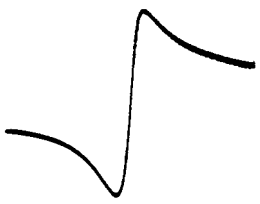


FIG. 9. Photograph of a dispersion signal from protons in water.

The outstanding feature of this method of determining magnetic moments is its very high precision, limited only by the accuracy with which resonance conditions can be ascertained, and illustrated by Figure 8, which represents an absorption curve. It would seem that exact resonance corresponds to the exact maximum of this absorption curve; this, however, would not be a safe conclusion since the position of the maximum is sensitive to slight but unknown disturbances. It is far safer to state that resonance occurs somewhere between the two points at half-maximum of the curve, and to accept its half-width as a measure for the experimental error. Since the experimental error amounts, in the case of Figure 8, to one-half oersted in a field of 2000 oersted, this curve would lend itself to the determination of the magnetic moment of the proton to one part in four thousand. This is, in fact, the approximate accuracy with which a great number of nuclear moments have been determined, and it is in most cases perfectly sufficient for the interpretation of their values in terms of nuclear structures.

There are, however, cases where higher resolution can give significant information, particularly where one is dealing with isotopes of the same

element, that is, with nuclei which differ in the number of neutrons which they contain but have the same number of protons. The ratio of the magnetic moments of two isotopes, if known with very high precision and compared to the results of other experiments, can lead to direct conclusions about the distribution of amperian currents within the nucleus.

For such cases, as well as for other purposes to be discussed later, it is desirable to obtain still sharper resonance curves. The width of such a curve is partly "natural," determined by the physical and chemical nature of the test substance, and partly "instrumental." One of the main causes of instrumental width is the variation of the field of the magnet over the sample region and this width can be reduced by improving the homogeneity of this field. Particularly in liquids there are many cases where the natural line-width is exceedingly small, so that the observed line-width depends primarily upon the care exerted to obtain very homogeneous fields. As an example of what has already been achieved, we shall give the results in the case of water. It has been possible to obtain in fields of 7000 oersted proton resonance curves with a line width of as little as 1/1000 oersted, corresponding to a resolution of one part in seven millions. A comparable accuracy has been obtained in the ratio of the magnetic moments of the proton and the deuteron, the nucleus of heavy hydrogen which consists of a neutron and a proton. This isotope is of particular interest to nuclear physicists because of its simplicity, and the measurement of the ratio has contributed towards the understanding of its properties.

Natural Width and Structure of Resonance Lines

It was stated above that the natural width of resonance lines depends upon the substance in which nuclear magnetism is investigated. Actually it can be demonstrated to be caused by internal fields arising from neighboring atoms and molecules which have a perturbing influence upon the precession of a nuclear moment. Conversely it is possible to use the study of the observed line-widths as a means to obtain information about the molecular surroundings of nuclei. Such studies have been carefully carried out in solids, liquids, and gases and the results have been applied to problems concerning their constitution and molecular motion.

Instead of merely causing a broadening of the resonance lines it is also possible for the internal fields to result in a splitting into several components. Such structures of lines have been particularly investigated in crystals where they can be related to the arrangement of the constituting atoms. Two major causes of line-splitting have here been recognized. One of them is the interaction of the nuclear magnetic moment with that of neighboring nuclei; the different components of a line structure are here due to the modification of the external magnetic fields by a small additional field which arises from neighboring magnetic moments and

depends upon the different possible orientations of these moments. Another important, cause of line-splitting is the existence of inhomogeneous electric fields in the crystal; these fields can be seen to exert, an additional torque on a nucleus which depends upon the distribution of charges in its interior and this torque differs for different, orientation of the nucleus. It is thus possible to obtain information about, the deviation of the charge distribution from spherical symmetry, measured by what is tarmed the "electric quadrupole moment," of the nucleus.

Chemical Shift and Resonance Structure in Liquids

Other causes of line structures are known and some of those which appear in liquids will be discussed below. There exists, however, still another modification of resonance lines which establishes a close connection between nuclear magnetism and chemistry. It consists of the displacement of resonance lines if the same nucleus is observed in the same external magnetic field but in different chemical compounds. The explanation of this "chemical shift" concerns the electrons which circulate around the nucleus as indicated in Figure 1. These electrons can be said to have a shielding effect in the sense that the magnetic field, acting upon the nucleus is slightly reduced from the value which it would have if all the surrounding electrons could be removed. Considering that. the chemical binding between atoms is due to their outermost electrons, the "valency electrons," and that it involves a modification of their orbits, it becomes plausible that, this modification also affects the shielding effect and, hence, the magnetic field which acts upon the nucleus. Chemical shifts are usually of the order of one part in a thousand; with much higher resolutions being attainable, they can easily be detected, measured, and subjected to a quantitative interpretation.

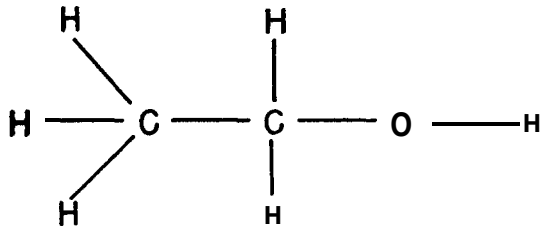


FIG. 10. Structure of the molecule of ethyl alcohol.

The existence of chemical shifts leads to a further question, namely, whether it is necessary to use different, chemical compounds for its observation or whether similar effects can occur within one and the same compound. Somewhat surprisingly it is found that the latter is the case, and we shall illustrate this "internal chemical shift" with reference to ethyl alcohol, with which most of the original work on this effect has been carried out. The molecule of this compound has the well-known structure shown in Figure 10. It indicates that three atoms of hydrogen (H) are tied to one carbon atom (C), two more hydrgen atoms to another carbon atom, and a fifth hydrogen atom to an atom of oxygen (O). The lines connecting various atoms represent "valency bonds" and sym-

boliae the fact that the binding between neighboring atoms is established by each one sharing a valency electron with another. The electrons in this structure constitute a different environment for the hydrogen atoms, bound in the three different groups (CH_3), (CH_2), and (OH); one may therefore expect that these electrons shield, by a different amount, the magnetic field acting upon the magnetic moments of the corresponding protons.

Figure 11 shows the actual existence of this internal chemical shift. It represents the photograph of the trace on the cathode-ray screen obtained from ethyl alcohol, in a manner similar to that of Figure 8 which was obtained from water. Instead of a single resonance curve there are three, originating from the protons in the CH_3, the CH_2, and the OH group in the molecule. The areas under these peaks are in the ratio 3 : 2 : 1, corresponding to the number of hydrogen atoms contained in each group.

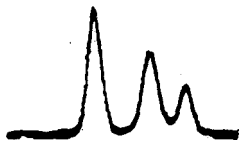


FIG. 11. Photograph of proton signals from ethyl alcohol. The peaks on the left, in the middle, and at the right are due to resonances in the CH_3, the CH_2, and the OH group respectively. The half-width of these resonance curves is equivalent to a variation of the magnetic field of about one part in a million.

The observation of this pattern was carried out in a field of about 7000 oersted and the separation between the maxima of the CH_3 and the CH_2 resonance curve was measured to be about 20 milli-oersted, or one part in 350,000. The relative line-width required to resolve the pattern must of course be even smaller-in this case in fact about one part in a million. Being almost entirely instrumental, i.e., determined by the variation of the field over the region of the sample, such a small line-width demands a correspondingly high degree of field homogeneity. This was achieved by careful shimming of the magnet and the use of a very small sample with linear dimensions of only a few millimeters.

It is evidently difficult to obtain still narrower lines by further improvement of the resolution and it might seem doubtful not only that it could be achieved but also that any more details would be revealed by this procedure. Figure 12 shows that both doubts are unjustified. It represents a duplication of the trace of Figure 11 with the difference, however, that it has been obtained under conditions where the resolution amounts to approximately one part in ten millions. As a' result of this very high resolution it appears that the three groups of hydrogen atoms which were discussed above lead not only to different resonance curves but that each gives rise to the existence of several closely spaced components.

The explanation of this further detail of structure is related to that found in many crystals. It originates from small contributions to the magnetic field, acting upon an individual proton, which arise from the

magnetic moment of other protons in neighboring groups of the molecule. The orientation of these moments with respect to the external field may be different in different molecules and thus leads to a multiplicity of effective field values and corresponding resonance lines.

While the splitting caused by the interaction of magnetic moments in crystals can amount to several oersteds, it is found in the case of ethyl alcohol to be of the order of a milli-oersted. This great reduction can be explained by the fact that unlike a crystal, where the position of the atoms remains essentially fixed, a liquid exhibits a molecular motion such that molecules change not only their position but at the same time carry out rapid random rotations around their centers of gravity. The field, due to the magnetic moment of a proton, is effectively averaged out by these rotations; but the small modification of the field by the valency electrons does not average out and thus gives rise to the observed structure.

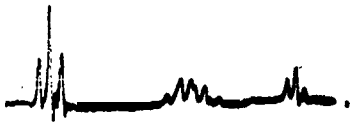


FIG. 12. Photograph of proton signals from ethyl alcohol under conditions of very high resolution. The three groups of lines represent the resolved structures of the three apparently single lines which have been obtained in Figure 11 under conditions of lower resolution. The half-width of the individual lines is here equivalent to a variation of the magnetic field of about one part in ten millions.

It can be seen that the mechanism leads to the observed order of magnitude of the splitting, but it would be very difficult to predict it numerically. There are, however, simple rules concerning the number of component lines in each group, which follow from the underlying mechanism and are experimentally verifiable. In this case, the rules indicate that where the splitting is due to other protons, each of them having a spin $\frac{1}{2}$, a group containing n protons causes the splitting of an otherwise single resonance line into $n + 1$ components of a neighboring group. Thus the CH_3 group, as well as the OH group, has the CH_2 group with two protons as its neighbor in the molecule and their resonances consist therefore of three components. The CH_2 group, on the other hand, has the CH_3 group with three protons as well as the OH group with one proton as neighbors. The former causes, therefore, a splitting into four components and the latter a further splitting of each of these into two more, resulting in a total splitting of the CH_2 group into eight components. Indeed, a close examination of the pattern in the center, arising from the CH_2 group, reveals that each of the three peaks in the middle of this pattern consists actually of two very closely spaced lines so that, together with the two small peaks on both sides, it is found to consist of eight components.

Structures similar to that of ethyl alcohol have been found in many other organic liquids, and molecules containing hydrogen and fluorine have been particularly investigated. The nuclei of fluorine have likewise

a spin ½ and their behavior is qualitatively the same as that of protons with the advantage, however, that the separations of the components of their resonance lines tend to be larger so that the requirements of resolution are less stringent.

While the detailed study of ethyl alcohol reported here has merely confirmed the structure of the molecule as it has been known for a long time, there are many molecules whose structure is not yet known. It is noteworthy that the number of components in the resonances of different groups of atoms is related to the character of neighboring groups. This aspect of nuclear magnetism is therefore capable of revealing neighborhood relations, thus furnishing an important clue to chemists in their attempts to establish and verify the structure of complex molecules.

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